

# Life history theory

## Learning objectives

1. Know the terms and reasons for analyzing structured populations
2. Convert life history information into a matrix for a stage-structured population
3. Get population growth rate from a matrix

*Review of last lecture: environmental variability and how to deal with that modeled population growth with seed bank*

*assumed all individuals are same - in reality, different stages & ages within population*

*Silvertown talked about tradeoffs between growth, survival and fecundity, we'll talk about this more next week built his analyses with structured population models, so introducing that first this week*

## Structured populations

- structured populations: accounting for stage or age of individuals *stage examples: seeds, vegetative, reproductive*

*question: any ideas why it might be useful to incorporate this information for a population?*

- differences in mortality more realistic

*example: differences in mortality, seedlings die in frost, reproductive adults don't*

- individual stages - consequences for population dynamics

*population growth in a population of only seedlings will look different than in population of older adults*

## Terminology

- demography - study of structured populations
- vital/demographic rates - germination, survival, fecundity *rates that describe each stage*

*vital & demographic used interchangeably*

- age and stage *related but useful in different circumstances*
- stable age/stage distribution - proportion of individuals in each age/stage

## Matrix population models

*let's say we have a plant with 3 stages: seed, seedling, and adult*

*we're interested in getting germination rates, survival rates, fecundity*

*question: any ideas how might we actually collect that data in a real population with a bunch of individuals in seed, seedling, adult stage?*

- data collection: counts in each stage, tracking individuals
- rates of transition between each stage

*no seed bank, all seeds are coming from adults in previous year*

| Year | seeds | seedlings | adults |
|------|-------|-----------|--------|
| 1    | 10    | 10        | 10     |
| 2    | 40    | 1 + 2     | 8 + 3  |

*in year one, we tag seeds, seedlings, and adults, and measure fecundity of adults*

*in year two, we have seeds from last year's adults, 1 previously tagged seedling, 2 new seedlings, 8 previously tagged adults, 3 new adults*

*quick aside: has this population reached its stable stage distribution? in other words, are the percentages of seeds, seedlings, and adults consistent from one year to the next?*

| Year | seeds | seedlings | adults |
|------|-------|-----------|--------|
| 1    | 33%   | 33%       | 33%    |
| 2    | 74%   | 6%        | 20%    |

- initial population  $\neq$  long term dynamics
- transient period
- number of individuals not constant, proportion in each stage constant

*returning to matrix population model*

*this information from 1 year transition is all we need to create a matrix population model, assuming stable stage distribution*

- each column = stage
- column 1 = rates from seeds to seeds/seedlings/adults
- column 2 = seedlings to seeds/seedlings/adults
- column 3 = adults
- adults to seeds is fecundity

```
stage <- c(0,0.2,0,0,0.1,0.3,4,0,0.8)
matrix(stage,nrow=3,ncol=3)
```

```
##      [,1] [,2] [,3]
## [1,] 0.0 0.0 4.0
## [2,] 0.2 0.1 0.0
## [3,] 0.0 0.3 0.8
```

*label columns & rows*

*can think of 0s as percentage of one stage going to others*

*for example, [3,1] is 0 because no seeds are going straight to adults*

*[1,2] is 0 because no seedlings are going back to seeds*

## Calculating $\lambda$

*use linear algebra, not going to get into that, just some definitions*

- eigenvalue = a scalar used in matrix multiplication
- several eigenvalues for a given matrix *i.e.* several scalars can work when doing this transformation/matrix multiplication
- largest, positive eigenvalue of matrix =  $\lambda$  = population growth rate

*really easy to do in R, will do in lab*

## (maybe skip) Age-structured populations: Leslie matrix

*let's put this in the leslie matrix format*

- key difference: can't stay in an age *like you can in a stage*
- fecundity in first row
- each age is a column
- survival to next stage on the diagonal

```
leslie <- c(0,0.2,0,0,0,0.25,100,0,0)
matrix(leslie,nrow=3,ncol=3)
```

```
##      [,1] [,2] [,3]
## [1,]  0.0 0.00 100
## [2,]  0.2 0.00  0
## [3,]  0.0 0.25  0
```